

Generation Interconnection System Impact Study Report

For Queue Project Q14-0320

McCross 138 kV
October 20, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0320. The Interconnection Customer proposes a 175.0 MW solar photovoltaic facility interconnecting at the McCross 138 kV substation in the West Virginia area. The request was received on March 22, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 178 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,863,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

LOWMOOR 345-138 kV Autotransformer: 108.9% loading (359 MVA) following the outage of the CRANDALL – ASPEN 345 KV; DFAX = 5.1%.

For the loss of ANTELOPE VLY STATION – MEADOW BRIDGE 138 KV, the N TULLAHOMA – BATESVILLE 138 KV loads to 106.8% of its 180 MVA rating. Project DFAX: 13.1%.

GREENLAND GAP – SANTEE COOPER 345 KV: 100.6% loading (1,470 MVA) following the outage of the MORGANTOWN ENERGY FACILITY – PRIDES CORNER 345 KV; DFAX = 5.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LOWMOOR 345-138 kV Autotransformer	Replace transformer with higher-rated unit	\$15,406,000	6% (\$900,000)
N TULLAHOMA – BATESVILLE 138 KV	Reconductor line	\$3,118,000	29% (\$891,000)
GREENLAND GAP – SANTEE COOPER 345 KV	Reconductor line	\$2,897,000	2% (\$72,000)
	Total allocated to project		\$1,863,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.